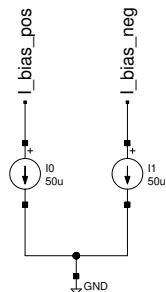
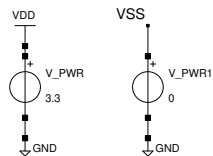
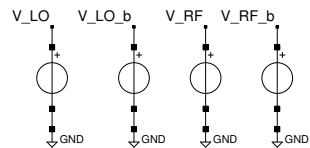
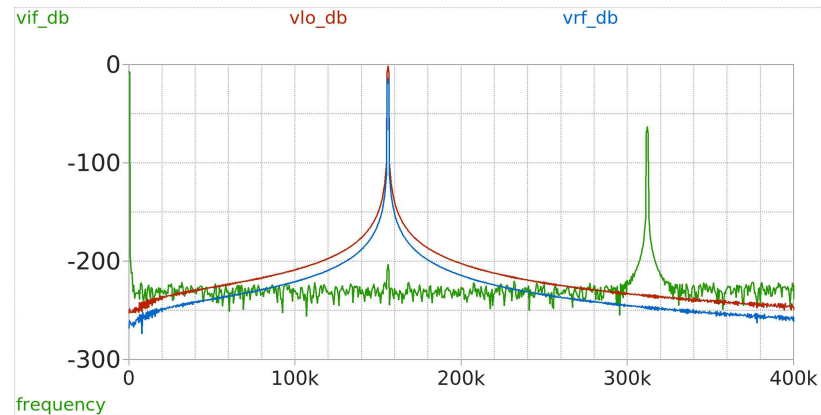
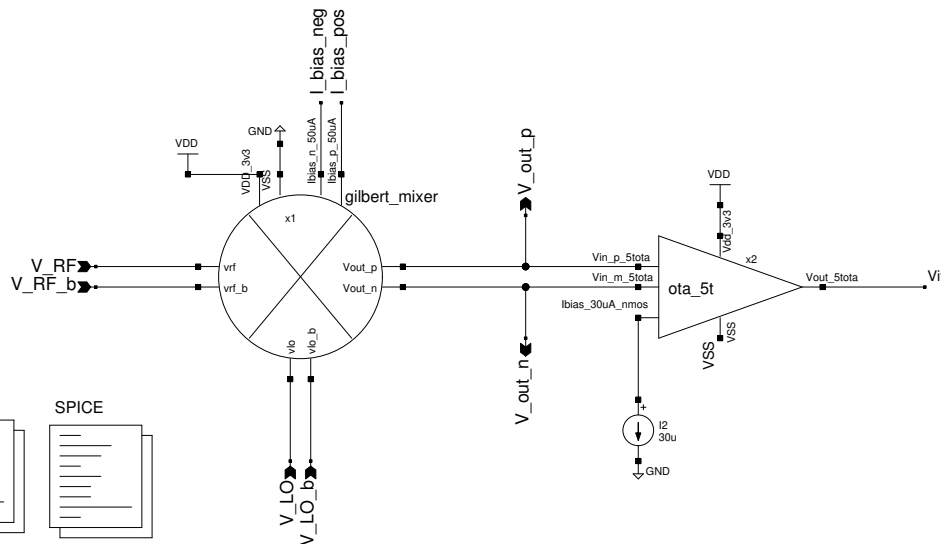


- ➡ Run ngSpice simulation (ctrl+left-click)
- ➡ Load ngSpice waveforms (ctrl+left-click)



MODELS

SPICE



COMMANDS

```
* let sets vectors to a plot, while set sets a variable, globally accessible
.control
```

```
* Set frequency and amplitude variables to proper values from within the
* sine-wave LO
* set cm_lo = 0.5
* set freq_lo = 2.50G
* set amp_lo = 0.5
* alter @V_LO[sin] = [ $cm_lo $amp_lo $freq_lo 0 ]
* alter @V_LO_b[sin] = [ $cm_lo $amp_lo $freq_lo 0 0 180 ]
```

```
set freq_lo = 156k
set cm_lo = 1.8
set amp_lo = 0.4

set cm_rf = 1.2
set freq_rf = 156k
* set freq_rf = 10.7Meg
set amp_rf = 0.1
```

```
* set the parameters to the voltage sources
* alter @V_LO[pulse] = [ 2 2.5 0 0.5p 0.5p 5n 10n ]
* alter @V_LO_b[pulse] = [ 2 2.5 5n 0.5p 0.5p 5n 10n ]
alter @V_LO[sin] = [ $cm_lo $amp_lo $freq_lo 0 ]
alter @V_LO_b[sin] = [ $cm_lo $amp_lo $freq_lo 0 0 180 ]
alter @V_RF[sin] = [ $cm_rf $amp_rf $freq_rf 0 ]
alter @V_RF_b[sin] = [ $cm_rf $amp_rf $freq_rf 0 0 180 ]
```

```
save all
```

```
tran 1000p 2m
```

```
let if_out = v(v_out_p) - v(v_out_n)
let vlo = v(v_lo) - v(v_lo_b)
let vrf = v(v_rf) - v(v_rf_b)
```

```
plot vlo vrf v(if_out) v(vif) ylimit -1 3.3
```

```
linearize v(vif) vlo vrf
```

```
fft v(vif) vlo vrf
```

```
let vlo_db = db(mag(vlo))
let vrf_db = db(mag(vrf))
let vif_db = db(mag(v(vif)))
```

```
plot vlo_db vrf_db vif_db xlimit 0 400k
```

```
write /workspace/Analog/schematics/2_mixer/simulation_files/mixer_tb.raw
```

```
.endc
```